Multi-Fault Load Protection System: Design and LTspice Simulation of a Voltage- and Current-Based Disconnect Circuit

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Abstract

This report presents the design, control logic, and LTspice simulation of a compact load-protection circuit that disconnects the load whenever the supply voltage or load current drifts outside a safe operating range. The circuit employs a behavioural source to model the relay/controller's decision-making logic and a voltage-controlled switch to model the contactor. Four fault conditions are considered: over-voltage, under-voltage, over-current, and under-current. The design was tested using a 50 Hz source, with its amplitude and equivalent load resistance varied in steps over time to sequentially simulate each fault condition. While the simulation confirms that the trip logic successfully disconnects the load, it also reveals two practical issues that we discuss openly: comparing instantaneous values against an AC sine wave causes the contactor to chatter near every zero-crossing, and upon opening, the load node voltage collapses to zero a state the controller itself misinterprets as a new under-voltage fault, causing the load to latch in the off state. We explain the underlying causes, demonstrate them via recorded waveforms, and outline the modifications required such as RMS/peak sensing, source-side sensing, a debounce/lockout timer, and a unit-scaling adjustment for current thresholds before implementing the concept in hardware.

I. INTRODUCTION

Electrical loads are frequently damaged by conditions they were not designed to withstand: sustained overvoltage stresses the insulation; severe undervoltage can stall or overheat a motor; excess current indicates a short circuit or overload; and low current may signal an open circuit or connection failure issues that should be identified rather than ignored. The assignment calls for the design of a control circuit utilizing a protection relay, a contactor/relay, a circuit breaker, and metering that maintains the load connection under normal conditions but automatically trips in response to any of these four abnormal states. At this stage of the project, we implemented and verified
the decision logic and switching action in LTspice before employing physical relays and circuit breakers. A behavioural voltage source (B1) acts as the protection relay: it continuously monitors load voltage and current, outputting a clear 1/0 control signal. A voltage-controlled switch, gated by this signal, functions as the contactor. This approach allows for the rapid testing of the entire protection system including thresholds, timing, and switching prior to any hardware implementation.

II. SYSTEM DESIGN

Fig. 1 (end of this section) shows the final schematic. It has three functional blocks.

A. Source and Load Emulation

Main_Source is a 50 Hz sinusoid, V = sin(2π·50·t), whose amplitude is switched between 311 V, 380 V and 150 V at scripted time breakpoints so that the same run exercises a normal supply, an over-voltage event and an under-voltage event in sequence (Table I). The current sink bi models the load: it draws I = V(load_node) / R(t), where R(t) is stepped between 440 Ω, 25 Ω and 5000 Ω to emulate a normal load, an over-current fault and an under-current fault. Driving the load from a divide-by-resistance expression, rather than a fixed resistor, made it easy to script all four fault scenarios into one transient run.

B. Switching Element

S2 is an LTspice voltage-controlled switch, .model my_switch sw ron=0.01 roff=1Meg vt=0.5 vh=0.1, standing in for the contactor. The low on-resistance (0.01 Ω) and high off-resistance (1 MΩ) approximate a closed and an open contact, and the 0.5 V threshold with 0.1 V of hysteresis gives the switch clean, glitch-free transitions in response to the digital-like 0/1 signal from the controller.

C. Controller / Protection Relay

B1 is a behavioral voltage source that implements the trip logic in a single expression (Section III). Its output directly drives the gate of S2, so B1 is effectively the protection relay and S2 is the contactor it operates.

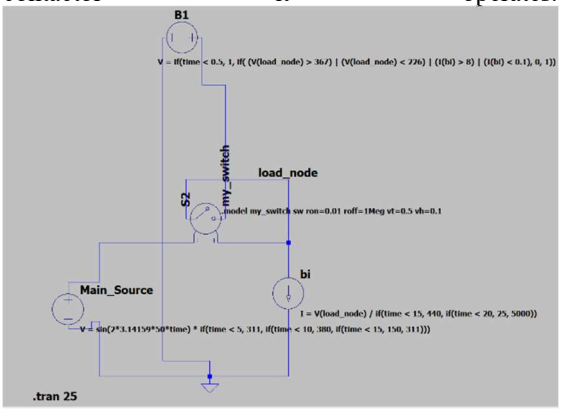


Fig. 1. Final circuit: 50 Hz source, load-current emulator (bi), voltage-controlled switch (S2), and the B1 behavioral protection-logic block that gates it.

III. PROTECTION LOGIC

The controller output is written as a single nested conditional in B1:

$$V(B1) = 1, \quad for\ t < 0.5\ s \quad (start\text{-}up\ hold) \qquad (1)$$

$$V(B1) = 0, \quad if\ V(load) > 367V \ \ or \ \ V(load) < 226V \ \ or \ \ I(bi) > 8A \ \ or \ \ I(bi) < 0.1A; \ \ else \ V(B1) = 1 \qquad (2)$$

The 0.5 s hold at start-up simply lets the source ramp up before the trip logic starts evaluating, avoiding a spurious trip on the very first instant of the run. After that, the load stays connected only while the load voltage sits inside roughly −27%/+18% of the 311 V nominal peak (220 V rms) supply and the sensed current sits between 0.1 A and 8 A; any single violation forces the output — and therefore the switch open.

One scaling issue is worth flagging here rather than hiding it: the current thresholds are written in amperes (8 A and 0.1 A), but I(bi) in this model only ever reaches the low-milliamp range (see Fig. 2). Because the simulated current never gets close to 0.1 A, the I(bi) < 0.1 A branch is true for the entire run, independent of the load condition being emulated. We discuss the consequence of this in Section V.

IV. SIMULATION SETUP

The circuit was run in LTspice as a 25 s transient analysis (.tran 25). Table I summarizes the scripted test schedule used to walk

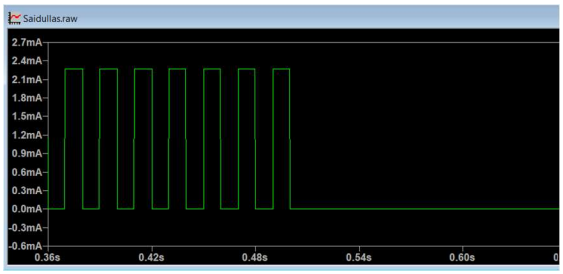


Fig. 2. Load voltage V(load node) and sensed current I(bi) for the final design. The load is chopped near each zero-crossing and then latches at 0V well before the scheduled fault windows are reached.

Two effects combine to produce this result, and neither is the over-/under-voltage fault the schedule was trying to test at that time. First, V(load) is compared to the thresholds instantaneously, not as an RMS or peak value. A 311 V-amplitude sine wave spends part of every cycle below the 226 V lower threshold simply because it is crossing zero, so the controller sees a spurious under-voltage condition twice per cycle and chatters the switch open and shut. Second, once the switch does open, load_node has no other path to be driven and its voltage collapses toward 0 V — which is itself read as an under-voltage condition, so the controller has no way to decide the fault has cleared and the load stays latched off. The current-threshold scaling issue noted in Section III reinforces the samethe circuit through normal operation and each of the four fault types.

Time (s)	Emulated condition	Source ampl. (V)	R eq. (Ω)
0 – 5	Normal supply	311	440
5 – 10	Over-voltage	380	440
10 – 15	Under-voltage	150	440
15 – 20	Over-current	311	25
20 – 25	Under-current	311	5000

TABLE I. SCRIPTED TEST SCHEDULE (FINAL DESIGN)

V. RESULTS AND DISCUSSION

Fig. 2 plots the load voltage V(load_node) and the sensed current I(bi) for the final design over the first 6 s of the run — the window covering start-up and the intended “normal supply” segment. Instead of a clean 311 V sine wave, the trace shows the load being chopped in and out roughly every half-cycle for about 1.2 s, after which both traces flatten to zero for the remainder of the recorded window outcome for the current-based branch: since I(bi) never reaches 0.1 A, the under-current condition is effectively always asserted.

We compared this against an earlier iteration of the circuit that used voltage-based tripping only, with no current terms in the B1 expression (Fig. 3). It shows the same qualitative behavior: rapid switching for about one second, then a flat zero for the rest of the run.This is useful evidence that the chattering/latching problem is a property of comparing an instantaneous AC voltage to fixed thresholds and then feeding the decision back from a node the switch itself controls, rather than something introduced when the current-based branches were added

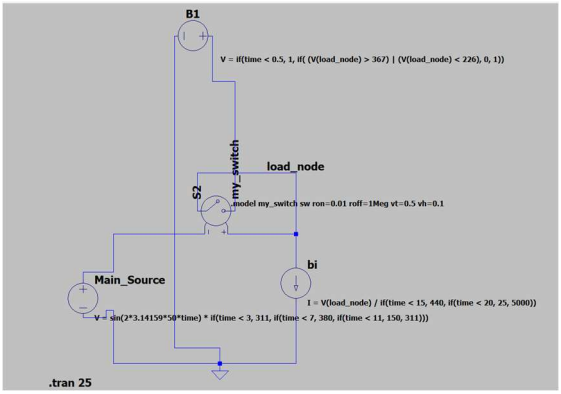


Fig. 3. Earlier iteration: voltage-only trip logic, no current terms in B1.

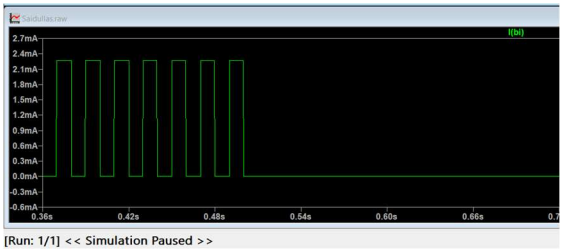


Fig. 4. I(bi) for the earlier voltage-only design, same chatter-then-latch pattern as the final design.

VI. LIMITATIONS AND FUTURE WORK

The simulation is a useful proof of the trip logic itself, but it is not yet a working protection scheme. Before moving to the relay/contactor/breaker hardware called for in the project brief, we plan to address:

1) Replace the instantaneous voltage comparison with an RMS estimate or a peak-detect-and-hold circuit so the controller reacts to the magnitude of the AC waveform rather than to every zero-crossing.

2) Sense voltage on the source side, or add a short debounce/lockout timer with an explicit reset condition, so the decision to re-close does not depend on a load_node voltage that only exists because the switch is already closed — the self-referential feedback is what causes the permanent latch-off once a trip occurs.

3) Correct the unit mismatch between the current thresholds (8 A / 0.1 A) and the simulated load current (sub-3 mA), either by rescaling the thresholds or by rescaling the load model, so the over-current path can actually be exercised and verified.

4) Carry the verified logic into the physical stage of the project: a real protection relay or comparator front end, a contactor or circuit breaker as the switching element, and a voltmeter/ammeter for live metering, broadly along the lines of standard protective-relay practice.

VII. CONCLUSION

We built and simulated a behavioral model of a multi-fault load protection circuit covering over-voltage, under-voltage, over-current and under-current conditions in LTspice, using a single behavioral source as the relay logic and a voltage-controlled switch as the contactor. The trip logic itself simulates as intended, but exercising it against a real AC waveform surfaced two concrete design issues — zero-crossing chatter from instantaneous thresholding, and a self-latching trip caused by sensing the same node the switch controls — plus a units mismatch on the current thresholds. Identifying these before building hardware is, in our view, the main value of the simulation stage, and Section VI lays out the specific changes we intend to make next.

REFERENCES

[1] W. H. Hayt, J. E. Kemmerly, and S. M. Durbin, Engineering Circuit Analysis, 8th ed. New York, NY, USA: McGraw-Hill, 2012.

[2] Analog Devices, Inc., LTspice XVII User's Guide, Norwood, MA, USA.

[3] IEEE Standard for Relays and Relay Systems Associated with Electric Power Apparatus, IEEE Std C37.90-2005.

[4] IEEE Standard Electrical Power System Device Function Numbers, Acronyms, and Contact Designations, IEEE Std C37.2-2008.